

Assessing Tract – Level Socioeconomic Inequalities in Walking Time to Shared Mobility Systems in Manhattan

Junxiang Gong
Cornell University, NY, USA

1. Introduction

One of the biggest transit systems in the United States can be found in New York City, particularly in Manhattan, which stands as the biggest hub for public mobility systems in terms of density and accessibility ([Federal Transit Administration, 2026](#)). Despite the abundance of transportation infrastructure, however, equal access is not assured within each tract. Among residents, racial and socioeconomic minorities are the groups that are most explicitly affected by such inequalities ([TransitCenter, 2021](#); [Baghestani et al., 2024](#)). [Lucas \(2012\)](#) frames this problem as transport-related social exclusion, in which mobility constraints can limit participation in urban life.

Therefore, inequality in accessibility hides behind Manhattan’s reputation as the biggest and most sustainable transportation center ([Arcadis, 2017](#)). Nevertheless, current studies do not successfully analyze transportation inequalities from a spatial approach ([Sy et al., 2021](#); [Goldberg, 2023](#); [Baghestani et al., 2024](#)). As for that, this study initiates with a core question: *To what extent do tract-level socioeconomic characteristics explain inequalities in public and shared mobility systems in Manhattan?* It is followed by three sets of questions and studies that concentrate on differences in travel time, spatial patterns of accessibility levels, and the relationship between accessibility and inequality.

Next, this study captures buses, subways, and shared bike systems, sourced from OpenStreetMap and combined with the American Community Survey (ACS) 5-year estimates and TIGER/Line shapefiles. The study first uses OSMnx tools to capture the nearest nodes, and measure the shortest travel time from census tracts in Manhattan to each node to assess how long it takes to access public and shared mobility systems. Next, it clusters the tracts using the K-Means Clustering algorithm and visualizes spatial patterns of accessibility time in the city. Third, spatial regressions are conducted to identify any relationship between socioeconomic minorities and accessibility levels.

Finally, this study demonstrates inequality in accessibility, while leaving room for further studies. This study shows that subways take longer time than buses and bike share systems, and that there are spatial differences among Lower Manhattan, East Harlem, and areas bordering the Hudson River and the East River. It also identifies that poverty rate is positively related to inaccessibility to mobility systems, while racial distribution appears to be nonsignificant, contradicting prior research studies ([Goldberg, 2023](#); [Baghestani et al., 2024](#)). It concludes by outlining its potential f

or use in other boroughs of New York City.

2. Background

The notion of equity has become crucial in the transportation field, as inequalities in socioeconomic factors, including income, employment, housing, education, poverty, and access to services, lead to disadvantage and inaccessibility to the transport system, thereby resulting in social exclusion (Lucas, 2012; Karner et al., 2025). Public transportation systems, such as subway and bus transit, also exhibit inequalities in accessibility among socioeconomic minorities (Liu & Duan, 2020; Liang et al., 2025). In addition, shared mobility systems, such as bike share in cities, also face inequalities in access and usage due to socioeconomic disparities (Fortes et al., 2024). These public and shared mobility systems, designed and provided for the public, are conversely facing inequalities that contradict their initial purpose.

New York City, United States, is not an exception to these inequalities. New York City, with the largest population in the United States and one of the highest annual population increases (United States Census Bureau, 2026), is the largest hub for all transportation systems. New York City reports the highest ridership for public transportation systems among all cities in the United States (Federal Transit Administration, 2026), and it is also ranked as having the highest sustainability in transportation systems (Arcadis, 2017). In addition, bike share in New York City has been prominent through the Citi Bike system, providing over 17 million trips annually and positioning itself as a competitive mobility option relative to other public transit (Cui et al., 2023).

Despite the abundance of public transportation systems, New York City explicitly illustrates transportation inequalities. Baghestani et al. (2024) demonstrate that accessibility to the subway is unequal among racial groups in the boroughs of New York City, with Queens being a leading borough for accessibility inequality for Black and Hispanic residents. Similarly, TransitCenter (2021) shows that accessibility to jobs via transit in Metro New York has been racially uneven despite the considerable presence of transport infrastructure in this area. In addition, the Office of the New York City Comptroller (2018) reported that seven of 27 Manhattan neighborhoods, approximately 26 percent, do not have a single subway station. These findings imply that inequalities in shared mobility systems are prevalent in New York City, and that smaller-scale studies are needed to effectively capture these inequalities in Manhattan.

Studies on shared mobility inequalities have been constantly studied in New York City, with consideration of racial and socioeconomic disparities in the use of public and shared mobility systems (Sy et al., 2021; Goldberg, 2023; Baghestani et al., 2024). However, there are limitations in prior research studies, as they concentrate on identifying significant relationships between disparities and accessibility and do not provide a spatial analysis of inequalities in New York City. In addition, while other studies measure the distance and time to shared mobility stations in temporal ter

ms (Jin et al., 2019; Cui et al., 2023), these analyses are not integrated with the socioeconomic features to spatially visualize the relationships and evaluate accessibility levels. Therefore, narrowing the gap among three approaches—measuring accessibility to public and shared mobility systems, identifying relationships among racial and socioeconomic inequalities, and visualizing the spatial patterns of inequalities—requires further study in New York City.

In summary, this study adds to existing literature by exploring inequalities in public and shared mobility systems through a spatial approach. This will differ from previous research by focusing on identifying spatial patterns within Manhattan before examining inequalities between these accessibility patterns and socioeconomic disparities.

3. Methods

3.1. Research Question

This study centers on one research question, which is followed by three hypotheses. The main research asks: *To what extent do tract-level socioeconomic characteristics explain inequalities in public and shared mobility systems in Manhattan?* This is followed by three sequential questions, which are answered with three hypotheses prior to conducting the research:

Q₁: How long does it take to arrive at nearest shared public mobility systems in Manhattan?

H₁: Subways would take less time to access than bus stops and bike shares, due to higher ridership.

Q₂: Are there any spatial patterns showing inequalities to access to public mobility systems?

H₂: Lower Manhattan (specifically Midtown) will show better accessibility than other districts.

Q₃: What are the socioeconomic factors that have relationships with such patterns or inequalities?

H₃: Socioeconomic minorities (both poverty and race) will face inequalities to accessibility.

3.2. Geographical Boundary

The study area is Manhattan (or New York County) in New York City, NY, USA. Surrounded by the Hudson River on the West and the East River on the East, Manhattan stands as the crux of New York City. It also locates itself as a central connector to other boroughs of the city (Bronx, Queens, Brooklyn, and Staten Island), and neighbors New Jersey to the West. Therefore, it is the central hub of transportation systems and demonstrates the significance and necessity of accessibility studies.

3.3. Data Collection

The data for socioeconomic characteristics were sourced from ACS 2020 5-Year Estimates. Data from New York County (Manhattan) was collected using its FIPS code 061, and at the census-tract level from 303 tracts in Manhattan. From the ACS, population density, White & Non-Hispanic population, population below the poverty level, commuters by personal drive, and commuters by public transit were collected and converted into percentages for ratio-based comparisons. Population density is logged due to its skewness, especially in urban areas ([Visée et al., 2024](#)). **Table 1** shows how the variables are labeled and measured by units.

Table 1

Descriptive Coding Summary of Independent Variables

Description	Code Label	Measurement Units
Logged Population Density	log_pop_density	Logged Population Count
Percentage White & Non-Hispanic	pct_white	Percentage (%)
Percentage Below Poverty Level	pct_pov	Percentage (%)
Percentage of Commuters by Drive	pct_drive	Percentage (%)
Percentage of Commuters by Public Transit	pct_publictransit	Percentage (%)

In addition, TIGER/Line 2020 tract shapefiles were also collected for a spatial visualization of accessibility in Manhattan. They were merged with the ACS data on the GEOID variable to add geometry to the ACS table. The analytical dataset, after merging and filtering, comprised 303 Manhattan tracts with geometries for spatial analyses.

Finally, for mobility and street network data, parcel data of local bus stops, subway entrances, and bike share stations were collected from OpenStreetMap. For the walking network, the base map of Manhattan was imported based on the walking network type, and walking speed for travel was assumed to be 5km/h for this study. In total, the data included 1,677 local bus stops, 842 subway entrances, and 702 bike share stations (**Table 2**).

Table 2

Descriptive Coding Summary of Mobility Systems

Description	Tags (Keys)	Tags (Values)	Total Count
Local Bus Stops	highway	bus_stop	1677
Subway Entrances	railway	subway_entrance	842
Bike Share Stations	amenity	bicycle_rental	702
Total Count			3221

In the earlier stage of the analysis, regional bus stops were also considered for the accessibility study. However, the number of regional bus stations in Manhattan on OpenStreetMap was too low (11 total) to draw meaningful inferences. In addition, these regional bus stops serve as stations for traveling outside of New York City and to other cities or states, and therefore do not relate to the

e local accessibility measurements. Therefore, this mode of transportation was dropped from the analysis.

3.4. Analysis

Prior to encountering the hypotheses, all mobility-amenity points were snapped to the nearest nodes on the OSMnx walking network. In addition, tract centroids in Manhattan were extracted from the TIGER tract polygons and then snapped to their respective nearest walking network nodes. Then, the Multi-source Dijkstra shortest-path algorithm was performed using travel time (*travel_time*) as the edge weight between all amenity nodes for each mode, thereby assigning the minimum walking time values from any amenity in that mode to every network node. The shortest walking time obtained by multi-source Dijkstra was assigned to the tract centroid, which was snapped to the same node, and then converted to minutes.

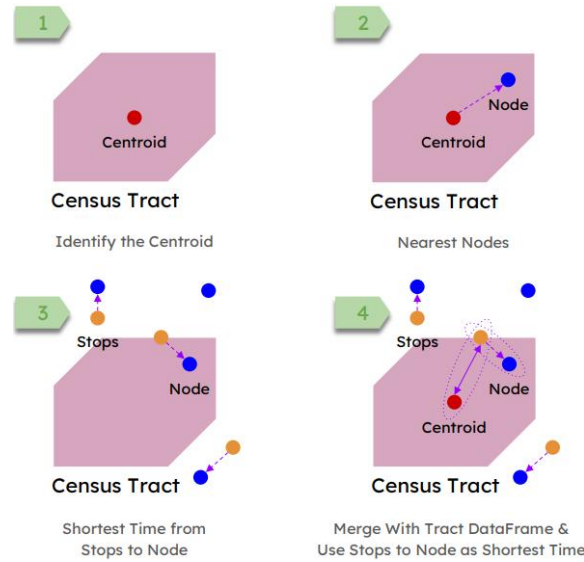


Fig 1. Measurement Steps for the Study.

After assigning the shortest time values to each tract, histograms for each of the three public and shared mobility systems were first depicted to visualize the distribution of the walking time among the 303 tracts. Choropleth maps were also created using quintile classifications. These maps helped assess the accessibility of each mobility mode, and the clustering of walking times on the map revealed the spatial distribution of long walking times across Manhattan.

Next, clustering analysis utilized the walking-time variables for local buses, subway entrances, and bike share stations as the feature vector for each tract. The preprocessing involved robust scaling of the three walking-time variables to minimize the effect of the right tail of the distribution. To determine the optimal number of clusters k , an elbow plot was generated for k values ranging from 1 to 9, showing the sum of squared errors. Based on the ideal number of clusters from the r

robust scaled data, K-Means clustering was performed, producing tract clusters and attaching cluster labels to each tract.

Finally, spatial regression analysis estimated one Ordinary Least Squares model for each mobility mode. K-nearest neighbors spatial weights matrix with $k=20$ was built based on the spatial location of each tract. Then, for each mobility mode (expressed as Y), spatial lag for Y was calculated as the mean walking time of the 20 nearest neighboring tracts. Therefore, all models were specified as the regression below:

$$Y \sim \text{spatial_lag_}Y + \log_pop_density + pct_drive + pct_publictransit + pct_white + pct_pov$$

4. Findings

4.1. Measuring the Time to Nearest Shared Public Mobility Systems

Prior to the study, Fig 2 shows the distribution of each mobility system in Manhattan, which are collected from OpenStreetMap. It shows that while local bus stops and bike share stations are evenly distributed across Manhattan, subway entrances are relatively clustered in Lower Manhattan, near Midtown and the Financial District. This implies that such spatial distributions may have an effect on inequality in accessibility among tracts, particularly for subway systems.

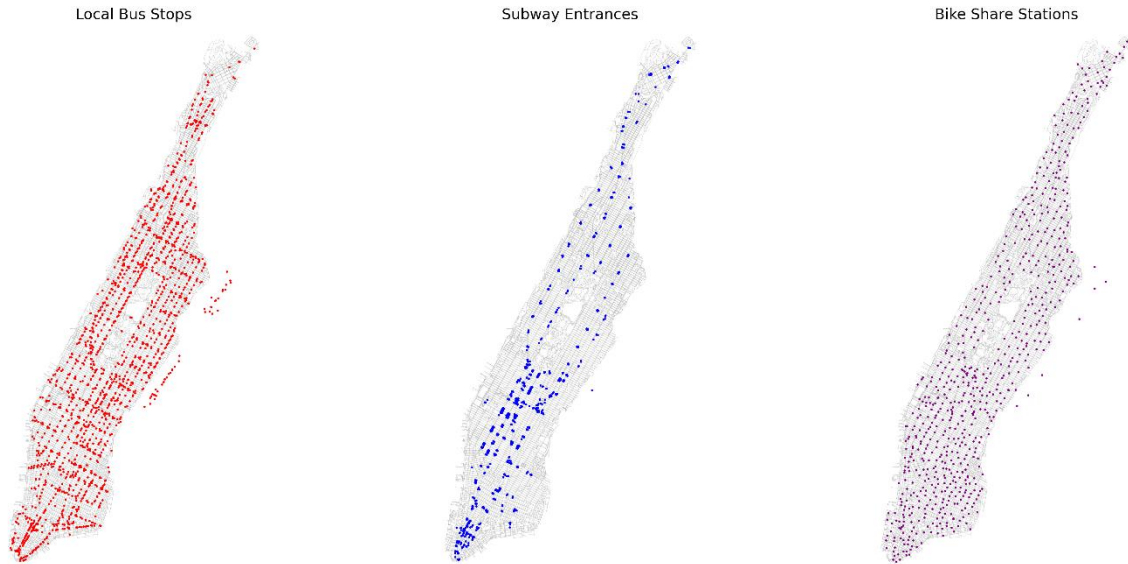


Fig 2. Distribution of Public and Shared Mobility Systems in Manhattan.

Then, this research begins the study by showing how walking times vary substantially across modes and census tracts in Manhattan. **Fig 3** shows that walking time to subway entrances mostly takes around 5 minutes, which is double the highest frequency of walking time to bus stops or bike share stations. In addition, **Fig 4** captures distinctive spatial patterns in walking-time gradients in

Manhattan. Lower Manhattan, specifically Midtown, constitutes the high-accessibility core of all transportation systems. In addition, while East Harlem appears to be highly accessible to local bus stops and bike share stations, it shows low accessibility to subway entrances. Finally, across all mobility systems, census tracts on river-facing edges appear to have the lowest accessibility, except for the Financial District in Lower Manhattan.

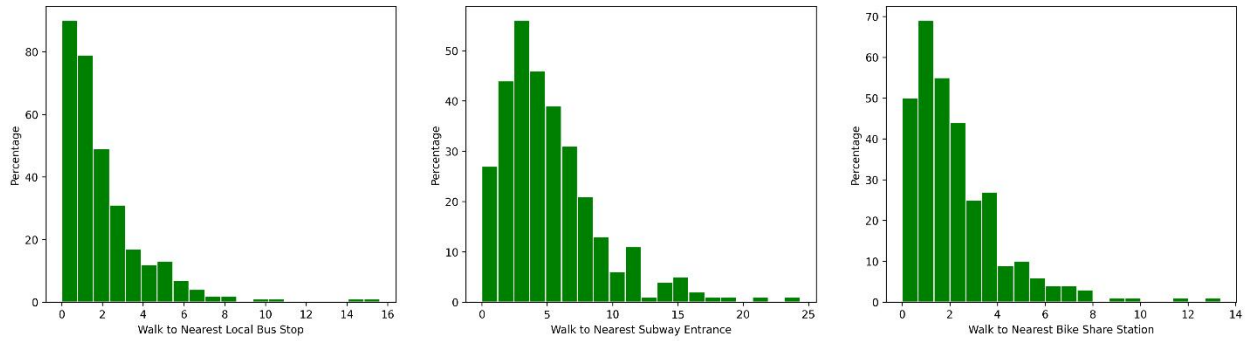


Fig 3. Histogram of Walking Time Variations to Nearest Public and Shared Mobility Systems.

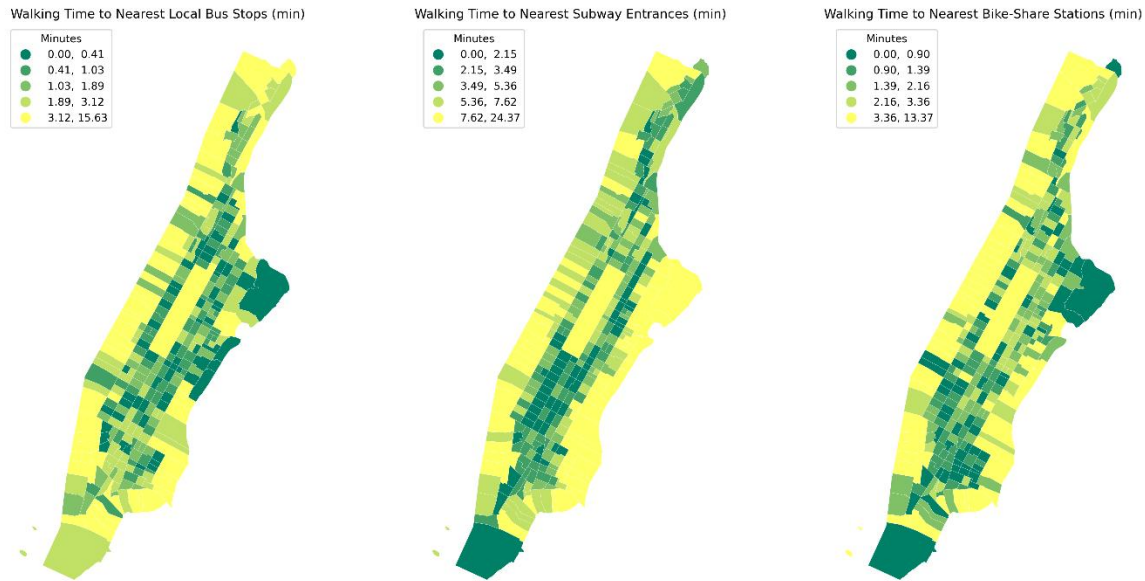


Fig 4. Walking Time to Nearest Mobility System, Illustrated as Quintiles

4.2. Clustering & Identifying Unequal Patterns

After the initial classification, K-Means clustering is applied to identify distinct accessibility typologies, with an aggregation of all three mobility systems. While previous maps independently show spatial patterns, they do not demonstrate a comprehensive spatial pattern. Therefore, the Elbow Method was used to identify the optimal number of clusters for Manhattan. Results showed significant drops in the SSE cluster 1 to 2, with a gradual decrease after 4. Therefore, four clusters

are selected as divisions for spatial patterns.

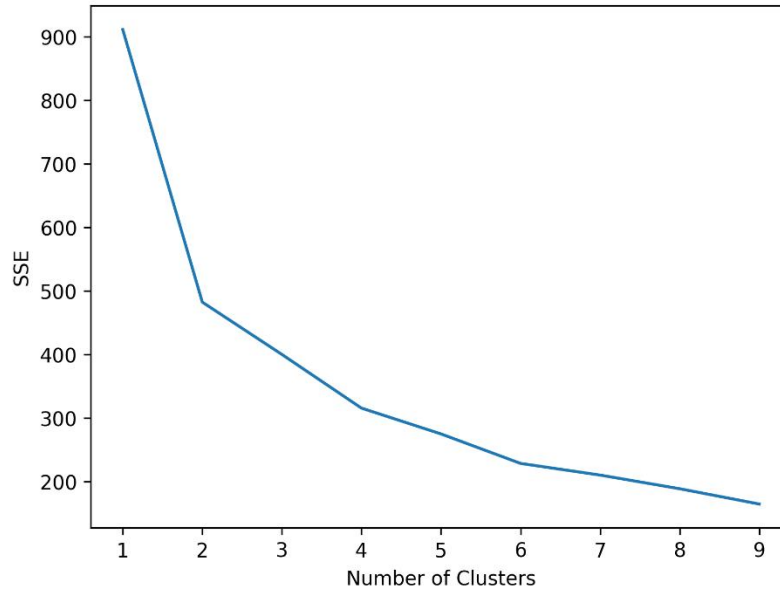


Fig 5. Elbow Method for Identifying the Ideal Number of Clusters

K-Means Clustering, using the Elbow Method, reveals four distinct accessibility typologies, as shown in **Fig 6**. Cluster 0 is mostly located at the outlying or river-edge tracts, which are the furthest tracts from the transportation modes. Cluster 1 is the most accessible area, with a mean walking time of 1.08 minutes to the nearest local bus stop, 2.65 minutes to the nearest subway station, and 1.41 minutes to the nearest bike share station. Lower Manhattan, particularly Midtown and the Financial District, makes up Cluster 1, while Cluster 2 comprises transitional tracts with mean walking times of 4.06 minutes to the nearest local bus stop, 7.99 minutes to the nearest subway station, and 4.25 minutes to the nearest bike share station. Therefore, each cluster can be classified as follows:

Cluster 0: Farthest to All Mobility Systems

Cluster 1: Closest to All Mobility Systems

Cluster 2: Moderately Far (Median) to All Mobility Systems

Cluster 3: Close to Bus and Bike, but Moderately Far to Subway

4.3. Spatial Regression Analysis

Finally, spatial regressions for all mobility systems are conducted, with travel time as the dependent variable. **Tables 3 ~ 5** below show the relationships between socioeconomic features and accessibility levels. Results show a positive relationship between poverty level and travel time to mobility systems, whereas race appears to be nonsignificant. In addition, the percentage of commuters who take public transit has a negative effect on travel time, while commuters who drive do not

have a significant effect on the relationship.

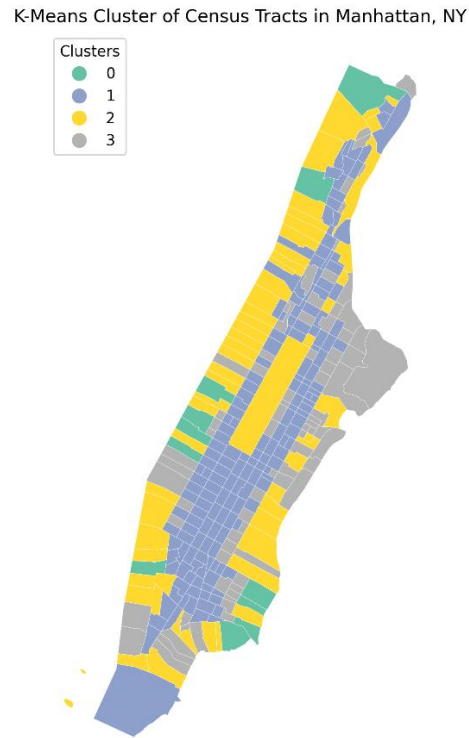


Fig 6. K-Means Accessibility Clusters of Census Tracts in Manhattan.

Table 3

OLS Regression for Buses

Travel Time to Local Bus Stop	Coef.	St.Err.	p-value	[95% Conf	Interval]	Sig
Spatial Lag	1.2468	0.137	0.000	0.000	0.978	***
Logged Population Density	-0.2364	0.120	0.050	-0.473	0.000	*
Percentage White & Non-Hispanic	0.5776	0.627	0.358	-0.656	1.811	
Percentage Below Poverty Level	3.2996	1.289	0.011	0.011	0.762	**
Percentage of Commuters by Drive	5.1373	2.686	0.057	-0.149	10.423	*
Percentage of Commuters by Public Transit	-1.6247	0.790	0.041	-3.180	-0.070	**
Adjusted R-squared	0.244	Number of obs		303		

*** $p < .01$, ** $p < .05$, * $p < .1$

Table 4

OLS Regression for Subways

Travel Time to Subway Entrance	Coef.	St.Err.	p-value	[95% Conf	Interval]	Sig
Spatial Lag	1.8313	0.1115	0.000	1.605	2.058	***
Logged Population Density	-0.5587	0.193	0.004	-0.939	-0.179	***
Percentage White & Non-Hispanic	0.4797	0.999	0.631	-1.486	2.446	
Percentage Below Poverty Level	5.3825	2.084	0.010	1.282	9.483	**
Percentage of Commuters by Drive	9.1455	4.254	0.032	0.774	17.517	**
Percentage of Commuters by Public Transit	-3.2959	1.237	0.008	-5.731	-0.861	***

Adjusted R-squared	0.502	Number of obs	303
*** $p < .01$, ** $p < .05$, * $p < .1$			

Table 5
OLS Regression for Bike Shares

Travel Time to Bike Share Station	Coef.	St.Err.	p-value	[95% Conf Interval]	Sig
Spatial Lag	1.1674	0.167	0.000	0.840 1.495	***
Logged Population Density	-0.2037	0.118	0.084	-0.435 0.028	*
Percentage White & Non-Hispanic	-0.1808	0.608	0.766	-1.378 1.016	
Percentage Below Poverty Level	0.5419	1.254	0.666	-1.926 3.010	
Percentage of Commuters by Drive	2.1478	2.628	0.414	-3.024 7.320	
Percentage of Commuters by Public Transit	-0.7370	0.776	0.343	-2.265 0.791	
Adjusted R-squared	0.137	Number of obs	303		
*** $p < .01$, ** $p < .05$, * $p < .1$					

5. Discussion

Prior to the analysis, this study set three different hypotheses to evaluate accessibility inequalities in Manhattan. Contrary to the initial assumption, it is shown that subway stations take a longer time to access than other public and shared mobility systems. This collides with the social preference that subways are used most frequently in New York City ([Federal Transit Administration, 2026](#)), and suggests that buses or bike shares may be a better option. However, this suggestion may not be flawless, as the total time to the nearest subway is mostly less than 10 minutes, which is still extremely walkable ([Maheshwari et al., 2025](#); [Yasin et al., 2025](#)).

In addition, spatial patterns document the second hypothesis that Lower Manhattan, particularly near Midtown and the Financial District, will have faster access to mobility systems than other tracts in Manhattan. However, this finding cannot lead to ascertaining that such regions do have higher accessibility levels, as this study solely focused on travel time. While time could be a measurement unit for accessibility inequality, studies on other transportation features should be conducted, as faster access does not guarantee a higher quality of service, such as frequency or crowdedness ([Ma et al., 2026](#)).

The limitations on the second hypothesis connect to the results and limitations of the third hypothesis. The three spatial regressions indicate travel time to each mobility system as the dependent variable; however, time is not the only factor that affects accessibility. There are various temporal, spatial, and environmental features that constitute accessibility that are not collected and measured in this study. This constraint stems from the data source, OpenStreetMap, and the methodology, OSMnx, which mainly extracts and evaluates access through travel distance and time. Therefore, determining which measurements to include and how to incorporate them into a single dependent accessibility index will be a crucial step in complementing this study.

There are also existing limitations in the methods of analysis. In the spatial regression, while poverty rate seemed to have a positive relationship with travel time, race did not appear to have a significant correlation for all three mobility systems. However, this is contrary to what previous researchers have conducted analysis on in New York City ([Goldberg, 2023](#); [Baghestani et al., 2024](#)), as studies utilize different datasets, variables, and research methods. Therefore, spatial regression should not be the sole method for evaluating relationships among features, especially in the urban field, where such relationships are intricately intertwined. Ultimately, while this study offers several clues that may support accessibility assessment, it also contains a few limitations in accessibility terminology and methods that need further investigation.

6. Conclusion

Every urban feature in a city cannot be ideally utopian: while the plethora of public and shared mobility systems in Manhattan makes the city the best transportation hub in the United States ([Cui et al., 2023](#); [Federal Transit Administration, 2026](#)), it also explicitly exhibits uneven distributions and unequal accessibility to such infrastructure ([Office of the New York City Comptroller, 2018](#); [Sy et al., 2021](#); [Goldberg, 2023](#)). It is therefore essential to profoundly consider accessibility and equity in the city, to reduce the hidden drawbacks and allow all residents in New York City to experience the highly reputable transportation systems.

This research, therefore, explored whether tract-level socioeconomic attributes contribute to accessibility inequalities in public and shared mobility systems in Manhattan at the census tract level. Local bus stops and bike share stations were evenly distributed, while subway entrances were concentrated in Lower Manhattan, resulting in longer travel times than those of the other two mobility systems. Such distributions also showed that it is easier to access public and shared mobility systems in Lower Manhattan, whereas areas near the rivers had lower accessibility. Finally, certain urban features, including the poverty rate and the percentage of commuters using public transit in a tract, were correlated with travel time to mobility systems, whereas racial composition was not significant.

However, the terminology of accessibility should constantly be investigated, as urban relationships are not connected in a singular relationship, but are intertwined with diverse intricate features, such as the quality of service, frequency of arrival, and crowdedness in public and shared mobility systems. This paper thus suggests one way to capture accessibility inequalities through a time lens, using spatial clusters to identify regions in Manhattan that are either privileged or discriminated against. The key, therefore, is recognizing the role of geographic distributions and socioeconomic determinants that could be aggregated to address a subjective urban feature in further research on transportation accessibility.

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